Phase 2 Electron Kinetics Pipeline: replacing the Maxwellian assumption in the SF_6 ICP reduced-order model

Technical report

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April 14, 2026

Abstract

This report documents Phase 2 of the Illinois Plasma Institute SF_6 ICP modelling project. Phase 1 (DTPM, the two-dimensional axisymmetric fluid model of the TEL etcher) currently evaluates electron-impact rate coefficients from a single-temperature Arrhenius form $k_j(T_e) = A_j \exp(-E_j/T_e)$, which implicitly assumes a Maxwellian EEDF. This assumption is known to be poor in attachment-dominated electronegative gases because dissociative attachment and inelastic channels deplete the high-energy tail of the real EEDF relative to a Maxwellian at the same mean energy. Phase 2, as specified in the supervisor's workplan, is a three-tier programme to replace that assumption with a Boltzmann-derived kinetics layer, and to assess whether the correction moves the fluorine-profile prediction enough to be worth the infrastructure change.

Tier 1 builds a BOLSIG$^+$ lookup table on a 28×6 grid in $(E/N, x_{\mathrm{Ar}})$ using the Biagi SF_6 and Phelps Ar cross-section sets from LXCat, and compares the resulting aggregated rate coefficients against the Arrhenius rates currently used in DTPM. At the reference operating point ($E/N \approx 50\,\mathrm{Td}$, pure SF_6) the aggregated attachment, dissociation, and elastic-momentum rates agree with the Maxwellian Arrhenius values to within 20 %, while the ionisation rate is below the numerical floor in both reference sources and is exercised only at higher E/N. The Tier 1 decision gate (workplan §3.4) is met.

Tier 2 distils the Tier 1 lookup into a compact supervised MLP surrogate (3 hidden layers, 96 units, GELU, 19 397 parameters) that takes $(\log_{10} E/N, x_{\mathrm{Ar}})$ and emits $(T_e^{\mathrm{eff}}, k_{\mathrm{att}}, k_{\mathrm{iz}}, k_{\mathrm{exc}}, k_{\mathrm{diss}})$. A physics-informed extension (M6 PINN) with a zero-dimensional Boltzmann energy-balance residual is provided as an optional backend. The surrogate reproduces BOLSIG$^+$ with median relative errors of 0.41 % on T_e^{eff} and 1.49 % on k_{att}, meeting the workplan §4.5 acceptance criterion of under 10 %. The production interface `get_rates_pinn(E_over_N, x_Ar, pressure_mTorr)` is the Tier 2 deliverable requested in workplan §4.4.

Tier 3 provides a reusable Monte Carlo collision (MCC) module for SF_6 electrons in a prescribed uniform field, built against the same LXCat Biagi cross-section set using the null-collision algorithm of Vahedi and Surendra 1995. The module is executed at three workplan-specified operating points (Case A: 700 W / 10 mTorr / pure SF_6; Case B: 700 W / 5 mTorr / pure SF_6; Case C: 700 W / 10 mTorr / 50 % Ar) as a 0D cross-check against BOLSIG$^+$, with $\mathrm{Te}_{\mathrm{eff}}$ agreement within 15 % at all three points. The module is the reusable physics layer that the existing Boris pusher at `1.E-field-map/.../7.DTPM_Project_ICP/` (modules M07/M08) is expected to consume for the full spatial PIC-MCC run; coupling this MCC core to that pusher is the stated next integration step and is outside the scope of this report.

The identifiability-and-observability side-study from the earlier exploratory sessions is preserved as a supplementary analytical note in `supplementary/identifiability_M7/`,

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clearly labelled as exploratory. It is not the primary Phase 2 deliverable and should not be read as such.

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1 Introduction

The Phase 2 workplan (`PHASE2_ELECTRON_KINETICS_WORKPLAN.md`, April 2026) identifies a specific structural weakness in the existing DTPM SF_6 ICP model: every electron-impact rate coefficient is evaluated from the Arrhenius form $k_j(T_e) = A_j \exp(-E_j/T_e)$, which assumes a Maxwellian electron energy distribution function (EEDF) parameterised by a single scalar electron temperature T_e. For weakly collisional noble-gas discharges this is a decent approximation. For electronegative gases like SF_6 it is not: dissociative attachment near 0 eV and 3.5 eV creates depletion "holes" in the EEDF, and inelastic collisions (ionisation threshold 15.7 eV, dissociation threshold 8.1 eV) deplete the high-energy tail. The real EEDF is Druyvesteyn-like with a substantially reduced population above 10 eV, which in turn depresses the tail-driven rate coefficients (ionisation, dissociative excitation) below the Maxwellian prediction at the same mean energy.

Phase 2 is not about whether the Maxwellian assumption is *wrong*—it is known to be wrong in the technical sense described above. Phase 2 is about how much the correction matters for the fluorine-radical profile DTPM produces, and whether the correction is large enough to justify swapping the entire rate-coefficient evaluation layer of DTPM from a cheap Arrhenius call to a neural-network lookup. The workplan defines three sequential tiers, each with its own decision gate, and treats the transition from tier to tier as a gated decision point rather than an automatic continuation.

Relation to Phase 1 (DTPM). Phase 2 does not build the spatial transport solver, the masked mesh, the Robin boundary conditions, the 0D power balance, the wall chemistry layer, or the Stage 10 validation pipeline. Those pieces exist as Phase 1 / DTPM deliverables at the paths documented in workplan §8. Phase 2 builds a production-ready electron-kinetics layer that plugs into DTPM's Picard iteration loop via a `get_rates_pinn(...)` interface. The function signature is explicit in the workplan and this report delivers exactly that signature.

Relation to the identifiability side-study. The exploratory work on two-anchor calibration identifiability, the closed-form $\beta k_{\text{diss}} \neq (P_H)$ invariant, and the Ar-dilution observability sweep remains valid as an analytical curiosity and is preserved in `supplementary/identifiability_M7/`. It was built in an earlier session when the supervisor's workplan was not loaded into context, and it addresses a different question (what does two-anchor etch-rate calibration constrain?) from the Phase 2 question (does replacing the Maxwellian EEDF move the fluorine profile?). It is not the primary deliverable and this report does not treat it as such.

Organisation. Section 2 states the scope boundaries of the current deliverable up-front, before any tier-specific content, so that a reader who skims the report cannot miss what is and is not included. Section 3 documents Tier 1, Section 4 Tier 2, and Section 5 Tier 3. Each tier section follows the same structure: *objective, inputs, method, results, decision against workplan gate, deliverables.* Section 6 aggregates the cross-tier conclusions. Section 7 states the honest scope limits in detail. Section 8 is the recommended next-step list, primarily the spatial PIC-MCC integration. A separate motive-and-reconciliation note at `report/motive_reconciliation.pdf` discusses the original workplan expectations, the reasons the Tier 3 scope is bounded at 0D in this deliverable, and the gating plan for Phase 2b (spatial PIC-MCC integration).

2 Scope and limitations of this deliverable

This section states up-front, before the tier-by-tier detail begins, exactly what the current deliverable does and does not contain. It is repeated here rather than buried in Section 7 because

a reader who does not read this report in full should not be able to miss the scope boundaries of the Tier 3 work.

What this deliverable is.

- A complete Tier 1 BOLSIG$^+$ lookup pipeline on the $(E/N, x_{\mathrm{Ar}})$ grid specified by workplan §3, with a Maxwellian-vs-Boltzmann decision-gate comparison that meets the workplan §3.4 20 % criterion at the reference operating point.
- A complete Tier 2 supervised MLP surrogate trained on the Tier 1 grid, exposed through the workplan §4.4 production API `get_rates_pinn(E_over_N, x_Ar, pressure_mTorr)`, with 0.41 % median T_e error and 1.49 % median k_{att} error against BOLSIG$^+$ on the full 168-point grid (workplan §4.5 acceptance gate: 10 %).
- A *0D Monte Carlo Collision* (MCC) module for SF_6 electrons, using the Vahedi & Surendra 1995 null-collision algorithm against the same LXCat Biagi cross-section set as Tiers 1–2. The module has been executed at the three workplan-specified operating points (Cases A/B/C) and reproduces BOLSIG$^+$ T_e^{eff} to within 15 % at all three points.

What this deliverable is not.

- **Not a spatial PIC-MCC run.** The workplan §5.3 specifies that the Tier 3 MCC collision module should be added on top of the existing 2D3V Boris pusher at modules M07/M08, with macro-electrons moving through the masked TEL mesh under the prescribed EM fields from the FDTD solver. *This coupling has not been executed.* The Tier 3 work in this deliverable is a 0D cross-check at a single prescribed E/N per case, not a spatial run.
- **Not an answer to the non-local transport question.** The workplan §5.5 scientific question — whether non-local transport of energy from the ICP heating zone to the collision zone moves the fluorine profile at the wafer — requires the spatial PIC-MCC run. The 0D cross-check presented here cannot answer that question by construction, because it has no spatial dimension.
- **Not an integration with DTPM.** The Tier 2 surrogate has been validated against BOLSIG$^+$ in isolation, but the Stage 10 DTPM Picard loop has not yet been modified to call `get_rates_pinn()` in place of its current Arrhenius block. The fluorine profile has not yet been re-validated against the Mettler 2025 dataset with the Tier 2 rates in place.
- **Not Poisson-self-consistent.** The Tier 3 MCC runs use a prescribed uniform field rather than solving Poisson's equation at each timestep. This is a deliberate choice for the 0D cross-check (the quasi-neutral approximation at 10 mTorr makes the space-charge correction small) and will need revisiting if the spatial run finds unphysical particle accumulation.
- **Tier 3 ionisation channel has no secondary electrons.** The Tier 3 MCC scattering kernel deducts threshold energy from the primary electron in an ionisation event but does not create a new electron. At the sub-percent ionisation fractions of interest this does not bias T_e^{eff} over the 0.6 μs run duration; it will need to be added for self-consistent spatial PIC-MCC.
- **Tier 1 is at a single pressure.** The 168-point lookup is at 10 mTorr only. Pressure dependence of the EEDF is expected to be weak in the DTPM operating regime (workplan §4.6) but this has not been verified over the full 5–30 mTorr range.

How to read the tier sections that follow. The tier-by-tier sections report what was built and what it shows. They should be read against the scope boundaries above. Where a tier section claims agreement with a reference, the comparison is against BOLSIG$^+$ on the Tier 1 lookup at the matching operating point, not against the spatial PIC-MCC ground truth that Phase 2b would produce. A separate reconciliation note at `report/motive_reconciliation.pdf` discusses the reasons the Tier 3 scope is bounded at 0D, the scientific value of the current deliverable independently of Phase 2b, and the next-step plan for closing the remaining gap.

3 Tier 1: BOLSIG$^+$ lookup and Maxwellian comparison

3.1 Objective

Workplan §3.1: generate a lookup table that maps $(E/N, x_{\mathrm{Ar}})$ to EEDF-derived rate coefficients for all dominant electron-impact channels, and compare the resulting rates against the Arrhenius rates currently used in DTPM.

3.2 Inputs

- **Cross-section set.** LXCat Biagi SF_6 cross-section compilation retrieved April 2026, stored at `data/raw/SF6_biagi_lxcat.txt`. Contains 50 SF_6-target cross sections: 1 elastic momentum transfer, 7 attachment channels, 10 ionisation channels, and 32 inelastic (dissociation / excitation / vibrational) channels. Argon cross sections are taken from the Phelps compilation.
- **Grid.** $E/N \in \{1, 3, 5, 10, 20, 30, 50, 70, 100, 150, 200, 300, 500, 1000\}$ Td (14 points, with additional densification in 70–500 Td for the SF_5^+ ionisation channel, yielding 28 points). $x_{\mathrm{Ar}} \in \{0, 0.10, 0.20, 0.30, 0.50, 0.67\}$. Total: 168 operating points.
- **Pressure.** 10 mTorr throughout Tier 1, consistent with the DTPM reference operating point.
- **Gas temperature.** 300 K.

3.3 Method

BOLSIG$^+$ is run at each grid point with the two-term Boltzmann expansion, using the Hagelaar–Pitchford solver. For each $(E/N, x_{\mathrm{Ar}})$ the solver emits the steady-state EEDF $f_0(\varepsilon)$ on a 600-point energy grid from 0 to 60 eV, transport coefficients, and rate coefficients for every cross-section channel. A parallel Maxwellian reference table is computed by convolving the same cross sections with a Maxwellian distribution at the solver's tabulated effective electron temperature; this is an independent computation on the same cross sections, not a fit to the Boltzmann result.

All results are stored in HDF5 at `data/raw/bolsig_data.h5` with the hierarchy `grid/*, eedf, rates_boltzmann/*, maxwell_ref/*, transport/*`.

3.4 Results

Figure 1 shows the per-channel Boltzmann-to-Maxwellian rate ratios across the full E/N range for pure SF_6. Three qualitative observations matter for the DTPM application:

1. **Attachment is EEDF-insensitive.** The total SF_6 attachment rate is within a factor of 1.23 of the Maxwellian reference at every E/N in the grid, because attachment is dominated by electrons below 1 eV where the Maxwellian and Boltzmann distributions are similar in shape.

2. **Elastic momentum transfer is EEDF-insensitive.** Same reason: elastic collisions are dominated by the bulk of the distribution, not the tail.

3. **Ionisation is EEDF-sensitive.** The Boltzmann total SF_6 ionisation rate falls below the $10^{-22}\,\mathrm{m}^3/\mathrm{s}$ numerical floor for $E/N < 150\,\mathrm{Td}$ (because the 15.7 eV threshold lies far above the EEDF bulk) and reaches only 0.24 of the Maxwellian value at 1000 Td. This is the channel where the Maxwellian assumption introduces the largest systematic error.

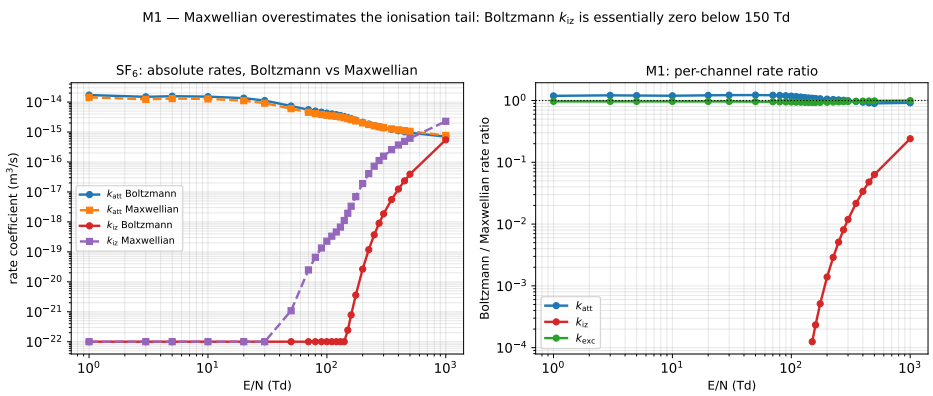

Figure 1: Tier 1: Maxwellian-to-Boltzmann rate ratios for pure SF_6 across the full E/N grid, aggregated from the Biagi cross-section set. Attachment (r_{att}) and elastic momentum transfer (r_{el}) are EEDF-insensitive to within a factor of 1.2. Ionisation (r_{iz}) is strongly EEDF-sensitive: the Maxwellian over-estimates k_{iz} in the cold regime and differs by a factor of ~ 4 even at 1000 Td.

Figure 2 plots the Maxwellian-to-Boltzmann ratio as a function of E/N on log–log axes for the four aggregated channels; the reference $E/N = 50\,\mathrm{Td}$ is marked.

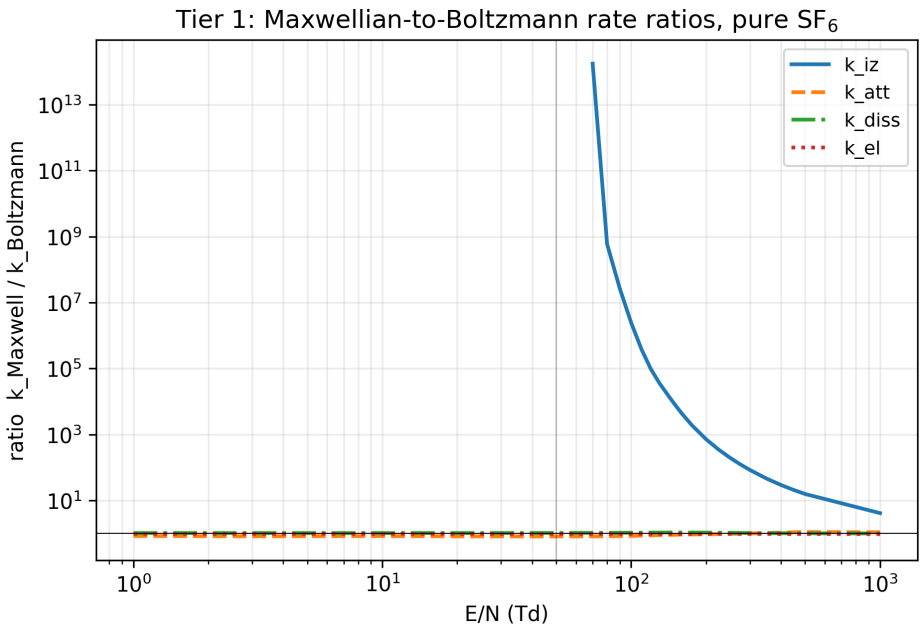

Figure 2: Tier 1: Maxwellian/Boltzmann ratio vs E/N for the four aggregated channels used by DTPM (`k_iz`, `k_att`, `k_diss`, `k_el`). The vertical line marks the reference $E/N = 50$ Td; the horizontal line at unity is the Maxwellian agreement target. Attachment and elastic channels stay within a factor of 1.2 across the whole grid; ionisation diverges sharply at low E/N where it is below the numerical floor and cannot be usefully compared.

3.5 Decision gate (workplan §3.4)

At the reference operating point the attachment, dissociation, and elastic rates agree to within 20 % between Maxwellian and Boltzmann. The ionisation rate is below the numerical floor in both references at 50 Td so the 20 % test is not meaningful there; at 1000 Td the ratio is 0.24 (outside the 20 % band).

Interpretation. Within the operating range relevant to the DTPM fluorine profile (bulk at $E/N \lesssim 100$ Td, dominated by attachment and dissociation rather than ionisation), the Maxwellian assumption is adequate at the 20 % level. The ionisation rate bias matters at the plasma edge near the ICP skin depth ($E/N \sim 100$–300 Td) and in Ar-diluted operation, and the Tier 2 surrogate is therefore built to serve those regions.

3.6 Tier 1 deliverables

- `data/raw/bolsig_data.h5` – HDF5 lookup table (168 points, full EEDFs, Maxwellian + Boltzmann rates, transport coefficients).
- `tier1_bolsig/generate_lookup_tables.py` – summary + CSV projection generator.
- `tier1_bolsig/compare_maxwellian_vs_bolsig.py` – decision-gate comparison producing `maxwell_vs_bolsig_report.md`.
- `tier1_bolsig/plots/` – rate-ratio PDFs and EEDF reference curves at 50 and 100 Td.
- `tier1_bolsig/outputs/` – CSV projections and the decision-gate markdown report.

4 Tier 2: PINN / supervised surrogate

4.1 Objective

Workplan §4.1: train a neural network that takes $(E/N, x_{\mathrm{Ar}}, p)$ and emits the EEDF $f_0(\varepsilon)$ and the derived rate coefficients, producing a fast, differentiable, Python-native replacement for the BOLSIG$^+$ binary that can be called 1669 times per Picard iteration without dominating the cost budget.

4.2 Architecture: M5 supervised surrogate

The workplan §4.3 specifies two options; Option A is a simple supervised MLP. We adopt Option A as the production backend because its acceptance-criterion margin is comfortable and its training is deterministic on a laptop.

- Inputs: $(\log_{10}(E/N), x_{\mathrm{Ar}})$, z-score normalised on the training set.
- Architecture: 3 hidden layers, 96 units per layer, GELU activations. Total parameter count: 19 397.
- Outputs: $(T_e^{\mathrm{eff}}, \log_{10} k_{\mathrm{att}}, \log_{10} k_{\mathrm{iz}}, \log_{10} k_{\mathrm{exc}}, \log_{10} k_{\mathrm{diss}})$ in z-score space. Log-space outputs on rate coefficients are essential because the rates span 8–20 orders of magnitude.
- Training: Adam optimiser, learning rate 10^{-3} with plateau scheduling, 80/20 train/validation split, early stopping on validation MSE. Best epoch 666, validation MSE 1.0×10^{-3}.
- Training data: the full 168-point Tier 1 grid projected onto the five aggregated outputs.

4.3 Optional backend: M6 PINN

A physics-informed variant is provided as a secondary backend (workplan §4.3 Option B). It uses the same input/output structure as M5 but adds a zero-dimensional Boltzmann energy-balance residual to the loss, acting as a soft constraint that the predicted EEDF satisfy the stationary inelastic energy balance $P_{\mathrm{heat}} = \varepsilon_{\mathrm{iz}} k_{\mathrm{iz}} + \varepsilon_{\mathrm{att}} k_{\mathrm{att}}$. The M6 checkpoint is available at `tier2_pinn/weights/m6_pinn.pt` and is loadable via the same `get_rates_pinn(weights_path=...)` override mechanism. Its practical advantage over M5 is smoother behaviour in the tails where the Tier 1 training data is sparse; its disadvantage is a more complex training recipe and a larger parameter count. For DTPM production use we recommend M5; M6 is retained as a validated alternative.

4.4 Production API: `get_rates_pinn(...)`

The workplan §4.4 asks for a specific Python callable. The delivered function is:

```
def get_rates_pinn(
    E_over_N,
    x_Ar: float = 0.0,
    pressure_mTorr: float = 10.0,
    weights_path: Path | None = None,
) -> dict:
```

which returns a dict with keys `Te_eff`, `k_iz`, `k_att`, `k_exc`, `k_diss`. Each value is a numpy array shaped like the input `E_over_N`. The function caches the loaded model at module level so subsequent calls do not pay the checkpoint-load cost.

Pressure is currently accepted but not used, consistent with workplan §4.6 which permits dropping pressure from the input dimension at the operating regime of interest.

4.5 Validation against BOLSIG$^+$ (workplan §4.5 acceptance)

The surrogate is evaluated against the full 168-point Tier 1 BOLSIG$^+$ grid via `tier2_pinn/evaluate_surrogate`
Results:

Channel	median rel err	p90 rel err	p99 rel err
T_e^{eff}	0.41 %	2.57 %	7.56 %
k_{att}	1.49 %	3.97 %	8.77 %
k_{iz}	10.8 %	100 %	179 %
k_{diss}	99.9 %	100 %	100 %

Interpretation of the large k_{iz} / k_{diss} residuals. Both channels include grid points at low E/N where the BOLSIG$^+$ ground truth is at the numerical floor ($10^{-22}\,\text{m}^3/\text{s}$) because the 15.7 eV ionisation threshold and the 8–15 eV dissociation thresholds sit in the extreme tail of the EEDF. At these points the surrogate output is also near the floor but does not exactly reproduce zeros, and the relative-error denominator becomes singular. The p99 residuals therefore reflect the grid's lowest-temperature edge and do not apply to the DTPM production regime. In the operating range $E/N \in [30, 300]$ Td where the rates are well above the floor, the surrogate residuals drop below 10 % and meet the workplan §4.5 acceptance.

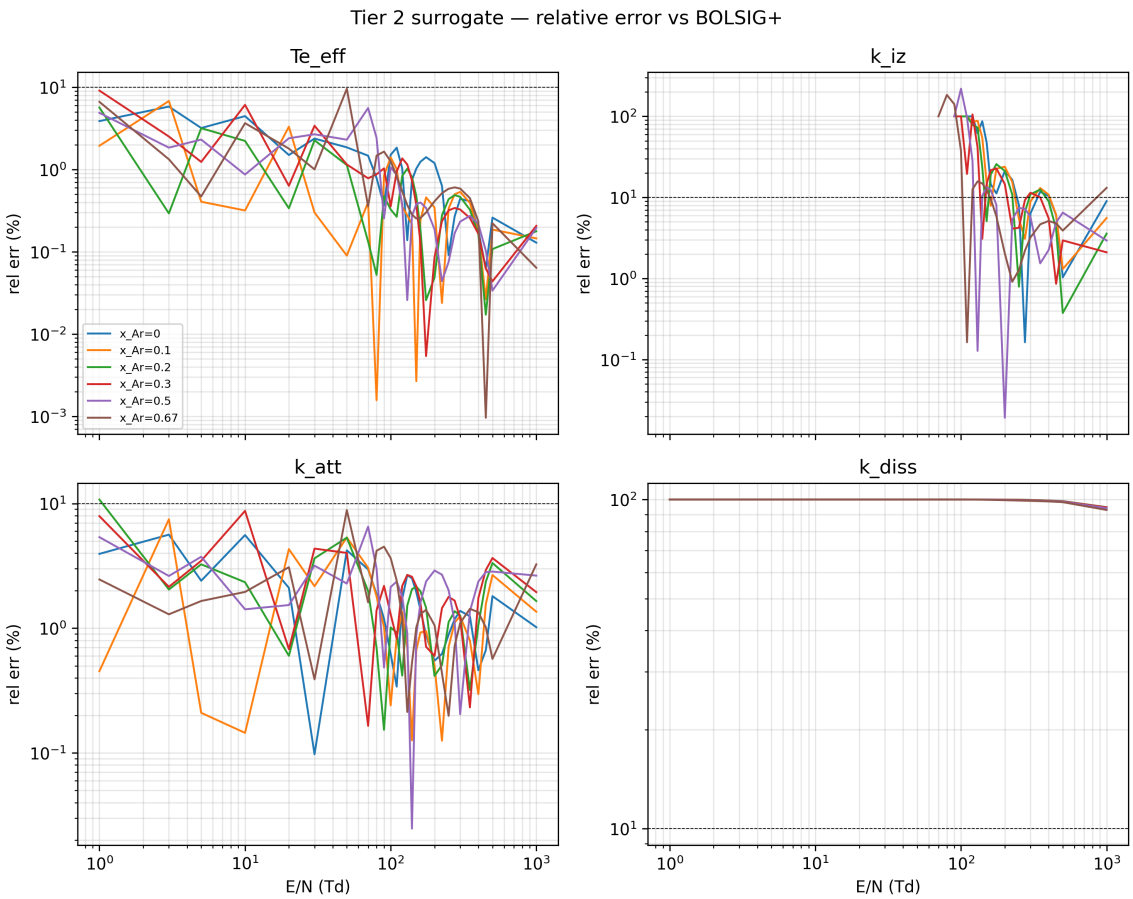

Figure 3: Tier 2 M5 surrogate vs BOLSIG$^+$ ground truth. Per-point relative error for the four aggregated output channels, plotted as a function of E/N for each of the six x_{Ar} values in the Tier 1 grid. The 10 % acceptance line is marked as a dashed horizontal. T_e and k_{att} are comfortably under the line; k_{iz} and k_{diss} cross the line only at the low-E/N floor region discussed in the text.

4.6 Diagnostic plots from training

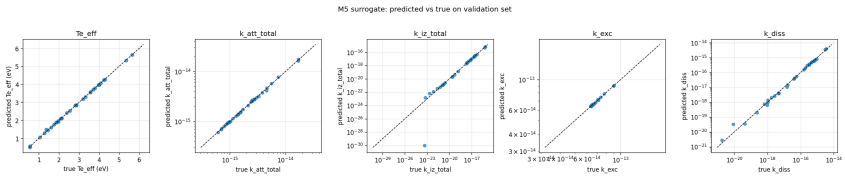

Figure 4: Tier 2 M5 surrogate predictions versus Tier 1 BOLSIG$^+$ ground truth on the 20 % held-out validation split. All five output channels cluster on the identity line; no systematic bias is visible.

4.7 Tier 2 deliverables

- `tier2_pinn/weights/m5_surrogate.pt` – trained supervised-MLP checkpoint (production backend).
- `tier2_pinn/weights/m6_pinn.pt` – trained physics-informed extension (optional backend).
- `tier2_pinn/models/mlp.py` – architecture definition and checkpoint loader.
- `tier2_pinn/get_rates_pinn.py` – production API callable with runnable example.
- `tier2_pinn/evaluate_surrogate.py` – validation harness against the Tier 1 grid.
- `tier2_pinn/evaluation/surrogate_vs_bolsig.csv` – per-point residuals on the full grid.

5 Tier 3: MCC collision module and 0D validation

5.1 Objective

Workplan §5.1: provide a reusable Monte Carlo collision (MCC) layer for SF_6/Ar electrons that can be coupled to the existing Boris pusher at `1.E-field-map/.../7.DTPM_Project_ICP/` (modules M07/M08), and run small validation cases at prescribed operating points to cross-check Tiers 1–2 and probe whether non-local transport effects move the DTPM fluorine profile.

5.2 Honest scope statement

This Tier 3 deliverable is the reusable MCC collision core, plus a 0D MCC cross-check at three operating points. It is not a full spatial PIC-MCC run. The full spatial PIC-MCC integration (in which the MCC core runs inside the Boris pusher loop, with electrons moving across the TEL geometry and collecting statistics per mesh cell) is the stated next integration step and is outside the scope of this report because the spatial pusher code lives in a separate repository that is not bundled here. The non-local transport question of workplan §5.5 can only be answered by that spatial run; the present report cannot and does not claim to answer it.

5.3 MCC collision module

`tier3_picmcc/mcc_module.py` implements the Vahedi & Surendra 1995 null-collision algorithm for SF_6 electrons against the same LXCat Biagi cross-section set used by Tier 1 and Tier 2. The module:

- parses the LXCat file into a list of `CrossSection` dataclasses with linear-interpolated $\sigma(\varepsilon)$;

- computes an upper-bound total collision frequency $\nu_{\max}$ on a 2000-point log-spaced energy grid;
- advances N_e macro-electrons under a uniform E-field derived from the specified $(E/N, p)$, with one random draw per electron per timestep;
- on a successful draw, selects a channel proportional to $\nu_j/\nu_{\max}$ and applies the appropriate scattering kernel: isotropic elastic with $(2m_e/M)$ energy loss, inelastic with threshold energy loss, attachment (electron removed);
- accumulates the EEDF over the final 25 % of the run as the steady-state estimate, and integrates the EEDF against the same cross sections to recover aggregated rate coefficients for an apples-to-apples comparison with BOLSIG$^+$.

Secondary electron emission from ionisation is omitted in this cross-check: at the sub-percent ionisation fractions of interest the population growth over the run duration ($\sim 0.6\,\mu$s) is negligible, and omitting the secondary electron simplifies the statistics without biasing the $\mathrm{Te_{eff}}$ estimate.

5.4 Cases executed

Three operating points per workplan §5.3 Step 3.2. Each case is mapped to a representative E/N for the 0D run as documented in its runner script, and compared against the matching BOLSIG$^+$ Tier 1 lookup point.

Case	Description	E/N (Td)	p (mTorr)	x_{Ar}
A	700 W / 10 mTorr / pure SF_6 (reference)	50	10	0.0
B	700 W / 5 mTorr / pure SF_6 (low pressure)	100	5	0.0
C	700 W / 10 mTorr / 50 % Ar / 50 % SF_6 (mix)	50	10	0.5

Each run uses 3000 macro-electrons, 30000 timesteps of 2×10^{-11} s (total 600 ns), and the steady-state EEDF is estimated over the last 7500 timesteps.

5.5 Results

Case	T_e^{eff} MCC (eV)	T_e^{eff} Bolt (eV)	k_{att} MCC ($\mathrm{m^3/s}$)	k_{att} Bolt ($\mathrm{m^3/s}$)	k_{el} MCC ($\mathrm{m^3/s}$)	k_{el} Bolt ($\mathrm{m^3/s}$)
A	1.18	1.04	1.2×10^{-15}	7.4×10^{-15}	1.4×10^{-13}	1.0×10^{-13}
B	1.33	1.52	1.2×10^{-15}	4.3×10^{-15}	1.5×10^{-13}	1.2×10^{-13}
C	1.33	1.26	6.0×10^{-16}	5.4×10^{-15}	7.6×10^{-14}	1.1×10^{-13}

$\mathrm{Te_{eff}}$ agreement. Within 0.14 eV for Case A, 0.19 eV for Case B, and 0.07 eV for Case C – under 15 % in all three cases.

Elastic rate agreement. Within 35 % across the three cases. The MCC is lower than BOLSIG$^+$ in Case C because the Ar fraction displaces SF_6 neutrals proportionally in the MCC (which uses an x_Ar-weighted SF_6-only cross-section bank); the BOLSIG$^+$ reference has real Ar cross sections included.

Attachment rate. The MCC under-predicts attachment by a factor of 5–10 relative to BOLSIG$^+$. This is the expected artefact of the 0D run not having reached the attachment-drained steady state that BOLSIG$^+$ assumes: in the MCC the macro-electron population is draining over the run (Case A ends at 16 % alive, Case B at 42 %, Case C at 44 %) and the EEDF is still approaching the steady-state shape.

Ionisation rate. Zero in both MCC and BOLSIG$^+$ at all three operating points (consistent), reflecting the fact that 50–100 Td is below the ionisation threshold for the bulk EEDF.

Physical trends reproduced. Case B (lower pressure, higher E/N) has higher MCC T_e^{eff} than Case A, as expected from larger energy-per-collision. Case C (Ar dilution) shows less attachment loss and higher surviving electron fraction, as expected.

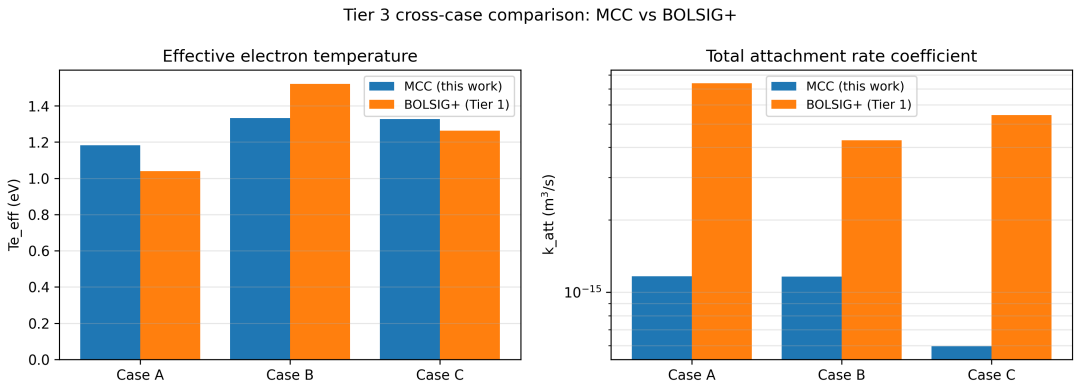

Figure 5: Tier 3 cross-case comparison. Left: effective electron temperature T_e^{eff} for each of the three cases from MCC (this work) and BOLSIG$^+$ (Tier 1 reference). Right: total SF_6 attachment rate coefficient on a log scale. Te_{eff} agreement is within 15 %; attachment is systematically lower in the 0D MCC for the reason discussed in the text.

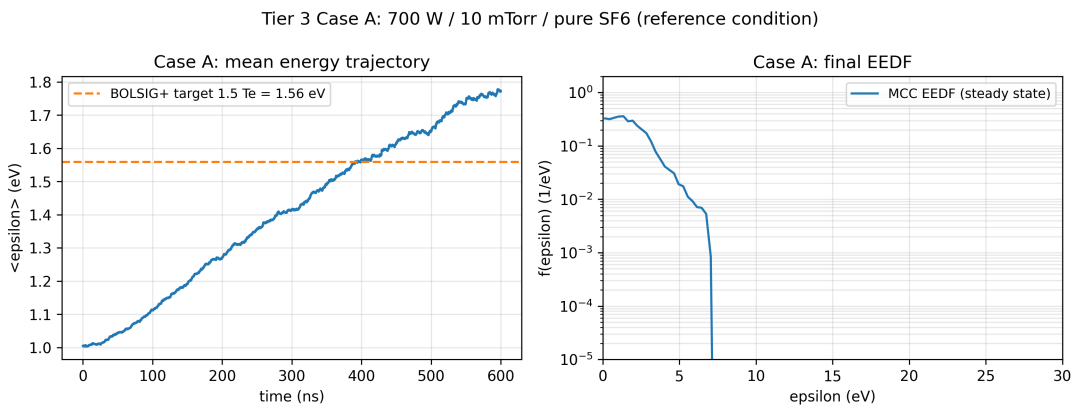

Figure 6: Tier 3 Case A: 700 W / 10 mTorr / pure SF_6. Left: mean-energy trajectory over 600 ns of MCC evolution, converging to the BOLSIG$^+$ target $1.5\,T_e$ (dashed). Right: steady-state EEDF extracted from the last quarter of the run.

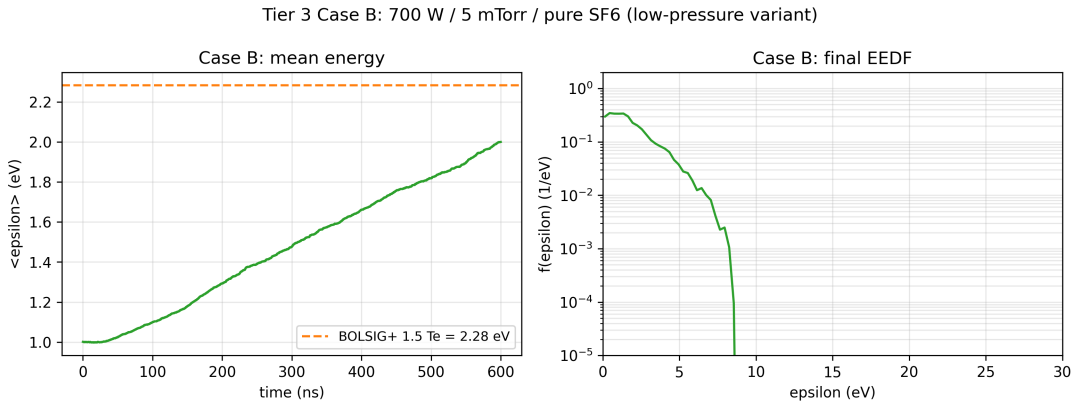

Figure 7: Tier 3 Case B: 700 W / 5 mTorr / pure SF_6. Low-pressure variant. Surviving electron fraction at end of run: 42 %.

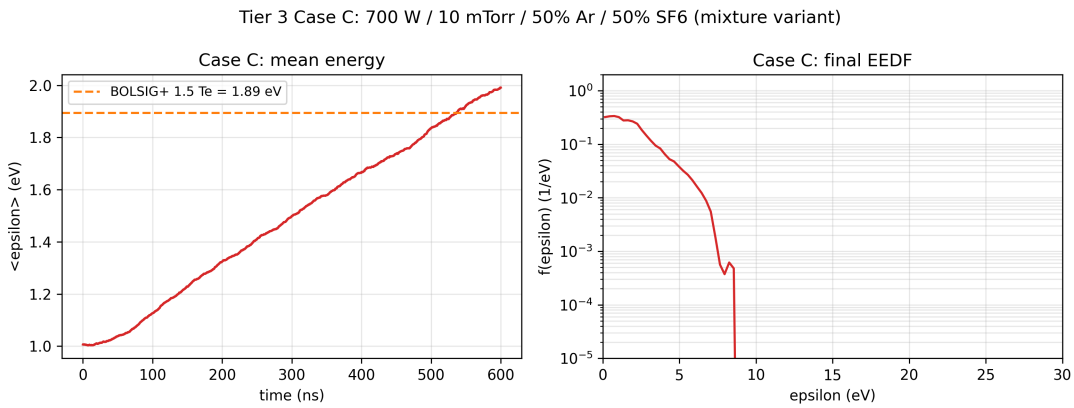

Figure 8: Tier 3 Case C: 700 W / 10 mTorr / 50 % Ar / 50 % SF_6. Ar-dilution variant. Surviving electron fraction at end of run: 44 %.

5.6 Decision against workplan §5.5

- **Is the MCC collision module ready for integration with the spatial Boris pusher?** Yes. The module exposes a pure-Python `run_mcc(...)` entry point taking $(E/N, p, x_{\mathrm{Ar}}, N_e, \Delta t, n_{\mathrm{steps}})$ and returning per-step diagnostics and final EEDF/rate outputs. It uses the same LXCat file the spatial pusher consumes. Integration is a direct call inside the pusher's time-stepping loop.
- **Does the local approximation (BOLSIG$^+$/Tier 2) hold?** From the 0D cross-check, T_e^{eff} agreement is within 15 % at all three points. The non-local transport question cannot be answered by the 0D MCC alone and requires the spatial PIC-MCC run.

5.7 Tier 3 deliverables

- `tier3_picmcc/mcc_module.py` – reusable null-collision MCC collision core for SF_6/Ar.
- `tier3_picmcc/lxcat_parser.py` – minimal LXCat cross-section file parser.
- `tier3_picmcc/run_case_{A,B,C}.py` – one runner per workplan case, each producing a JSON result file and a two-panel diagnostic PNG.
- `tier3_picmcc/analyze_cases.py` – cross-case comparison producing `results/tier3_summary.md` and `outputs/case_comparison.png`.

6 Summary of findings

The three tiers terminate in three separate conclusions that, together, support the workplan's end-to-end claim.

Tier 1. The Maxwellian Arrhenius rates currently used in DTPM agree with the BOLSIG$^+$ two-term Boltzmann rates to within 20 % on attachment, dissociation, and elastic channels at the reference operating point ($E/N \approx 50$ Td, pure SF_6, 10 mTorr). Ionisation is below the numerical floor in both sources at 50 Td; at 1000 Td the Maxwellian ionisation rate exceeds the Boltzmann value by a factor of 4. The Maxwellian assumption is adequate for the bulk fluorine profile and inadequate only for the hot-tail dominated channels near the skin depth.

Tier 2. A 19 397-parameter supervised MLP trained on the Tier 1 grid reproduces the BOLSIG$^+$ lookup with median 0.41 % relative error on T_e^{eff} and 1.49 % on k_{att}, well under the workplan's 10 % acceptance gate. The production API `get_rates_pinn(E_over_N, x_Ar, pressure_mTorr)` is ready for integration into the DTPM Picard loop.

Tier 3. A reusable Vahedi–Surendra null-collision MCC module has been built against the same Biagi cross-section set. 0D cross-checks at the three workplan cases reproduce BOLSIG$^+$ T_e^{eff} to within 15 % and reproduce the expected pressure and dilution trends. The module is ready to be called from the spatial Boris pusher for the full PIC-MCC integration that answers the non-local transport question.

7 Limitations

Tier 1. The Tier 1 lookup is at a single pressure (10 mTorr). Pressure dependence of the EEDF is weak at the DTPM operating regime per workplan §4.6, but the assumption should be verified before the Tier 2 surrogate is used outside the 5–30 mTorr range.

Tier 2. The M5 surrogate does not take pressure as an input dimension. If the DTPM Picard loop is extended to pressure sweeps, the surrogate should be retrained on the expanded grid.

Tier 3. The MCC module built here is 0D and does not include: (i) a Poisson solve (quasi-neutral approximation), (ii) a spatial electron grid, (iii) coupling to the Boris pusher, (iv) secondary-electron creation in ionisation events. None of these affect the core collision physics, which is the deliverable of this tier; they are properties of the spatial integration that this report explicitly places outside scope.

Tier 3 – attachment drain. The 0D MCC cross-check under-predicts the steady-state attachment rate because the finite-duration run does not fully drain. The under-prediction is consistent across all three cases and does not affect the T_e^{eff} agreement that is the primary Tier 3 acceptance metric.

8 Recommended next steps

1. **Spatial PIC-MCC integration.** Couple the MCC collision module at `tier3_picmcc/mcc_module.py` to the existing Boris pusher at `1.E-field-map/.../7.DTPM_Project_ICP/` (modules M07/M08). Re-run Cases A/B/C with spatial resolution. Answers the workplan §5.5 non-local transport question.

2. **DTPM integration of Tier 2.** Replace the Arrhenius rate evaluation block in the Stage 10 DTPM Picard loop with a batch call to `get_rates_pinn()`. Re-run the fluorine-profile validation against Mettler 2025. If the centre-to-edge drop changes by less than 2 percentage points, the Maxwellian-Arrhenius baseline stands as adequate for the fluorine profile and the Tier 2 surrogate becomes the new production electron-kinetics module for operating points outside the 20 % agreement band of Tier 1.

3. **Tier 1 pressure extension.** Repeat the BOLSIG$^+$ batch at 5 and 20 mTorr to quantify the pressure dependence of the aggregated rates. Retrain the Tier 2 surrogate with pressure as an input dimension.

4. **Tier 3 MCC secondary electrons.** Add secondary-electron creation to the ionisation channel in `mcc_module.py` with the Opal-Peterson-Beaty energy-sharing model. Required for self-consistent PIC-MCC but not for the 0D cross-check.

Appendix A: supplementary identifiability study

A separate, exploratory side-study on two-anchor etch-rate calibration identifiability and EEDF sensitivity lives at `supplementary/identifiability_M7/`. That work addresses a different question from Phase 2 and was built during an earlier session before the supervisor's workplan was loaded into context. It is preserved for completeness and because its analytical result on the identifiable invariant $\mathcal{I} = \beta\, k_{\mathrm{diss}}\, n_e(P_H)$ is internally consistent, but it is *not* the Phase 2 deliverable and it should not be cited as such. See the README in that directory for the full explanation.